Weekly Student Progress Performa

Team Id	CS26084	**Week no / Date**	Week 12 30 September 2026
Project Title	SentinelCore: A Network Detection and Incident Response Platform		
Members Present	Om Rashiya (24cs084), Yash Patel (24cs082)		
Mentor	Akshita Kadam		

WEEK 12 • Validation, Hardening, and Final Demonstration

Section	Details to be filled by the team
Planned Tasks (from last week)	Validate the complete platform, fix integration defects, polish documentation, and prepare the faculty demonstration.
Tasks Completed This Week	Ran backend, helper, pipeline, and browser-level checks; verified authentication, discovery, detection, search, incidents, reports, firewall, PCAP, and intelligence flows. Improved the README, test evidence, dashboard clarity, navigation, empty states, visual hierarchy, responsive behavior, and first-time-user guidance.
Commits Made	Total: 2 completion commits. 6945e77 — “updated and tested”; 19e4da8 — “Polish README with badges, TOC, and visual section dividers”. Later repository commits record maintenance, security hardening, and UI refinement.
Issues Created / Closed	Created: Integration testing exposed confusing empty states and several edge cases around parser output, reporting, and request security. Closed: Added clearer guidance, stronger validation, reproducible test documentation, and follow-up hardening tasks.
Milestone Status	100% of the planned SGP scope demonstrated. Core platform complete; subsequent commits are maintenance, security hardening, and UI refinement.
Blockers / Challenges Faced	Docker networking/capture behavior varies by host OS, and realistic packet data is environment-dependent. The demonstration uses health checks, bounded test fixtures, and

Research and Technical Study

Research objective. Validate the complete platform, fix integration defects, polish documentation, and prepare the faculty demonstration.

Study completed. Applied repository security guidance for SSRF, spoofed client IPs, rate limits, request validation, race prevention, and secure defaults. Also reviewed dashboard information hierarchy and progressive disclosure so a first-time user can understand system health and the next action immediately.

Final validation coverage

Validation area	Repository evidence	Outcome checked
Backend and RBAC	pytest and end-to-end API suite	Roles, happy paths, negative cases
Privileged helper	helper unit tests	Validation, guards, and allowed operations
Browser interface	browser-driven smoke suite	Login, navigation, forms, and errors
Pipeline and workers	module tests and fixtures	Ingestion, reports, PCAP, and recovery
Documentation	README and test report	Repeatable setup and disclosed limitations

Decision carried into the project

Ran backend, helper, pipeline, and browser-level checks; verified authentication, discovery, detection, search, incidents, reports, firewall, PCAP, and intelligence flows. Improved the README, test evidence, dashboard clarity, navigation, empty states, visual hierarchy, responsive behavior, and first-time-user guidance.

Implementation Record and Website Evidence

End-to-end coverage

Validated login and RBAC, asset discovery, sensor control, event ingestion/search, correlation, incidents, reporting, containment, PCAP analysis, threat intelligence, and browser navigation using repository test suites and fixtures.

Defects and hardening

Recorded and corrected integration issues affecting PCAP output, PDF report generation, logout/login timing, event facet caching, and repeated event requests; continued with request-security and worker-recovery hardening.

Documentation and usability

Improved the README, test evidence, dashboard status communication, role-aware navigation, empty states, responsive layouts, onboarding, and the final product presentation.

Final intelligence view

The Threat Intel workspace provides instant lookup, IOC/Matches/Sources tabs, type/severity filters, active-state filtering, add-IOC action, and columns for confidence, source, threat type, matches, and last seen.

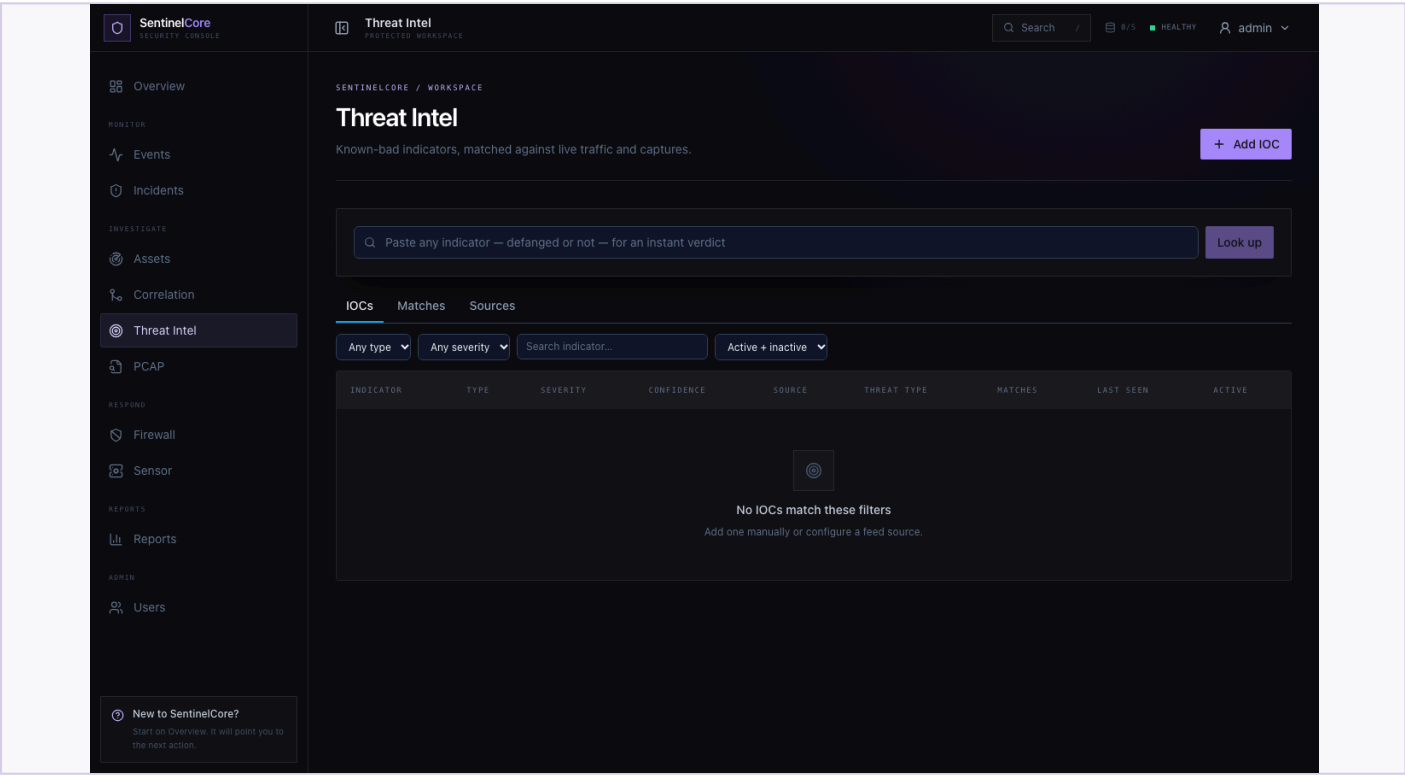


Figure W12.1 — Project-owned SentinelCore interface mapped specifically to Week 12.

Evidence interpretation. The final Threat Intel workspace shows the IOC lookup bar, IOC/Matches/Sources tabs, filters, Add IOC action, and the evidence columns used during investigation. Its visible controls, labels, status fields, and navigation provide direct evidence for the work described on this page.

Repository and Architecture Evidence

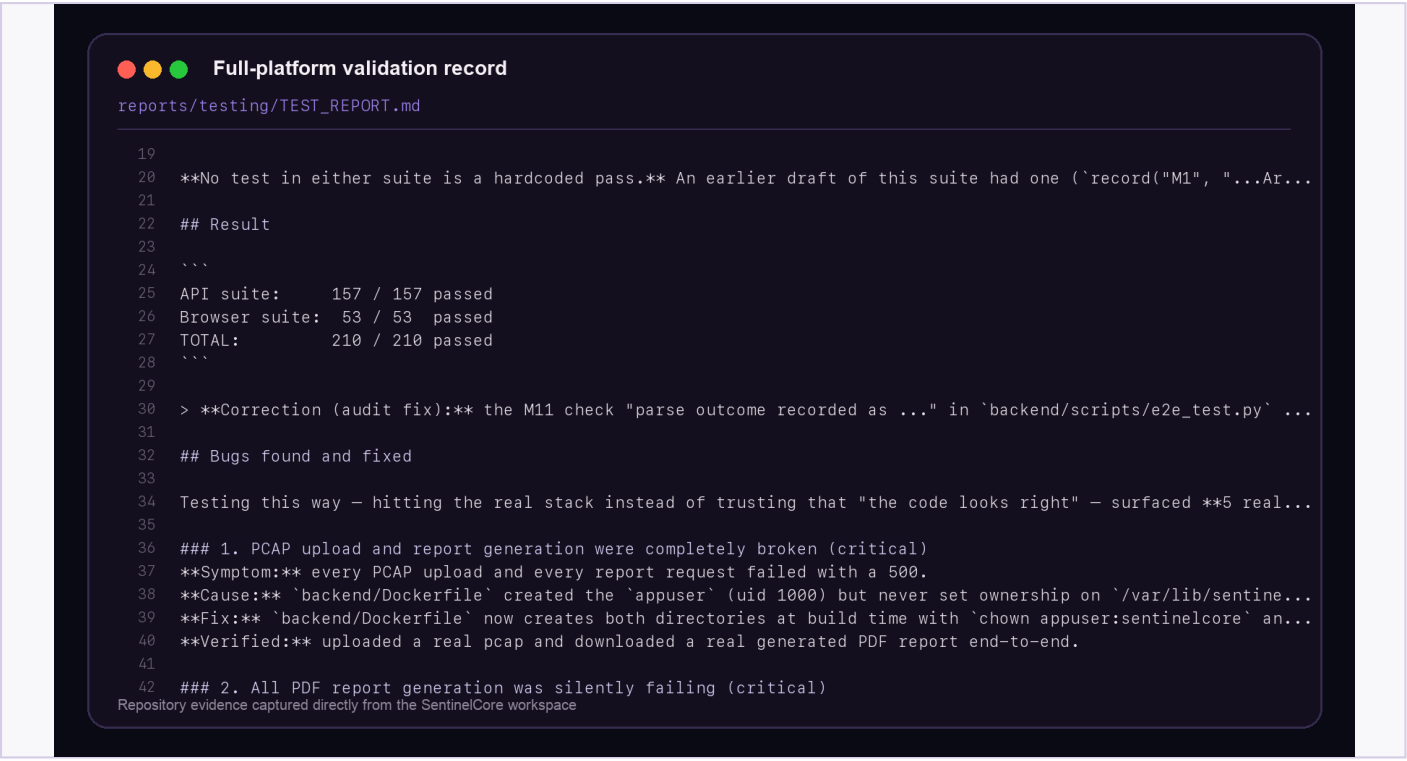


Figure W12.2 — Full-platform validation record. This is a separate internal artifact, not a copy of the website image.

Repository files used for traceability

- reports/testing/TEST_REPORT.md
- reports/validation/RESULTS.md
- README.md
- frontend/src/pages/intel/IntelPage.jsx
- frontend/src/pages/intel/SourcesTab.jsx

Verification

Completion claims were limited to the repository's demonstrated SGP scope and test documentation; later work is explicitly described as follow-up hardening rather than week-12 scope expansion.

Traceability conclusion. Full-platform validation record directly supports the Validation, Hardening, and Final Demonstration milestone. Completion claims were limited to the repository's demonstrated SGP scope and test documentation; later work is explicitly described as follow-up hardening rather than week-12 scope expansion. The artifact, source paths, implementation record, and website evidence therefore describe the same weekly scope.

Weekly Closure and Faculty Assessment

Completed outcome

Ran backend, helper, pipeline, and browser-level checks; verified authentication, discovery, detection, search, incidents, reports, firewall, PCAP, and intelligence flows. Improved the README, test evidence, dashboard clarity, navigation, empty states, visual hierarchy, responsive behavior, and first-time-user guidance.

Commit record

Total: 2 completion commits. 6945e77 — “updated and tested”; 19e4da8 — “Polish README with badges, TOC, and visual section dividers”. Later repository commits record maintenance, security hardening, and UI refinement.

Issue and resolution

Created: Integration testing exposed confusing empty states and several edge cases around parser output, reporting, and request security. Closed: Added clearer guidance, stronger validation, reproducible test documentation, and follow-up hardening tasks.

Milestone position

100% of the planned SGP scope demonstrated. Core platform complete; subsequent commits are maintenance, security hardening, and UI refinement.

Challenge handled

Docker networking/capture behavior varies by host OS, and realistic packet data is environment-dependent. The demonstration uses health checks, bounded test fixtures, and explicit setup guidance.

Next-week hand-off

Collect faculty feedback, continue security/performance hardening, expand realistic datasets, and prepare deployment documentation and final presentation material.

Repository reference

https://github.com/rashiyaom/SentinelCore

Assessed By (Faculty)	Marks Awarded (5 marks)	Faculty Remarks